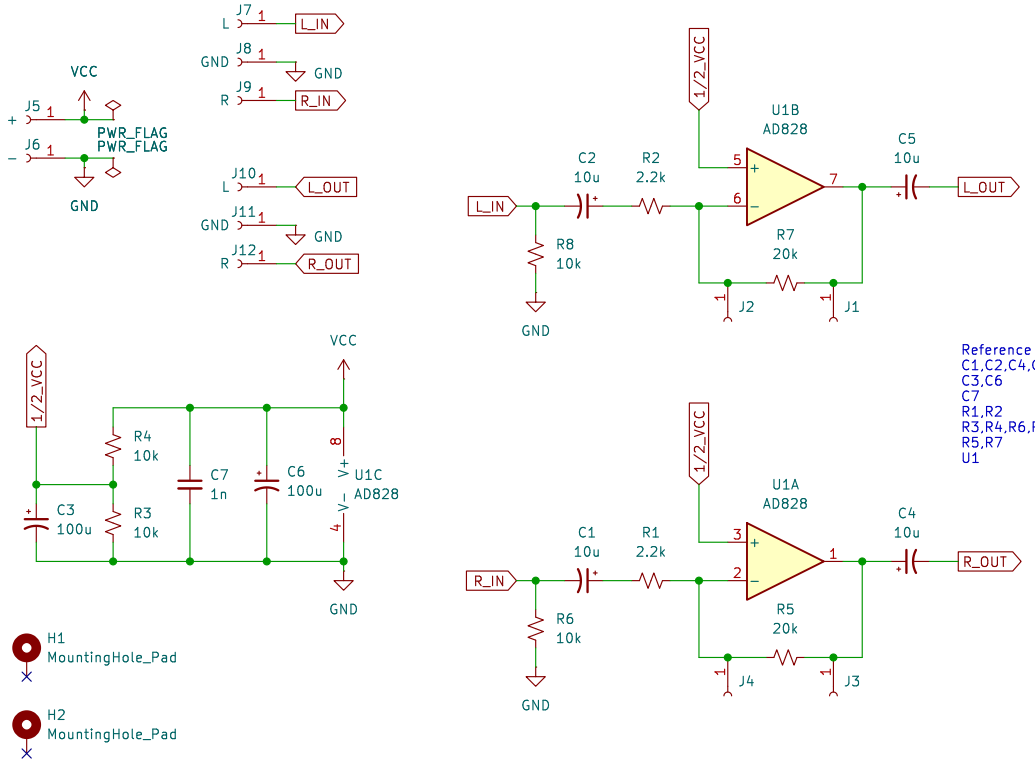
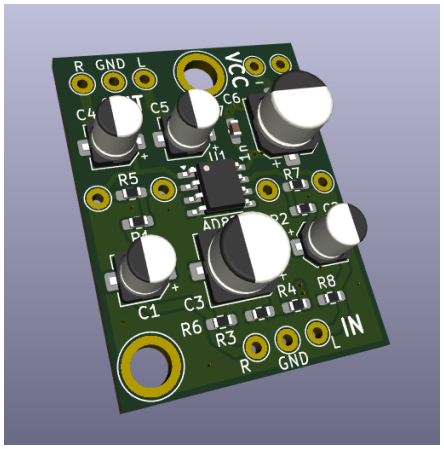




Reference	Qty	Value	Footprint
C1,C2,C4,C5	4	10u	CP_Elec_4x5.7
C3,C6	2	100u	CP_Elec_6.3x5.7
C7	1	1n	C_0603_1608Metric_Pad1.08x0.95mm
R1,R2	2	2.2k	C_0603_1608Metric_Pad1.08x0.95mm
R3,R4,R6,R8	4	10k	C_0603_1608Metric_Pad1.08x0.95mm
R5,R7	2	20k	C_0603_1608Metric_Pad1.08x0.95mm
U1	1	AD828	SOIC-8_3.9x4.9mm_P1.27mm

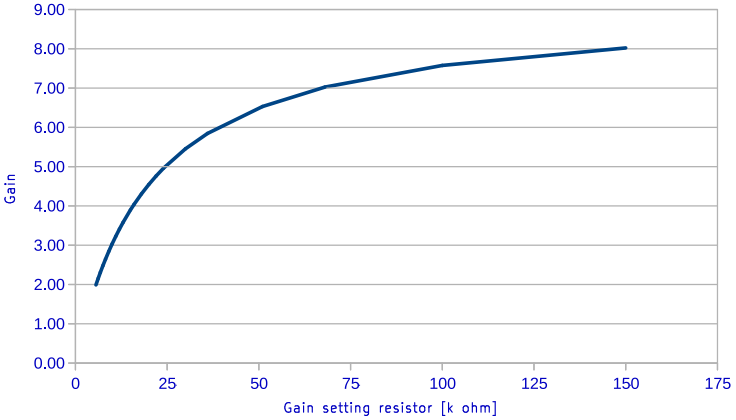
Power supply ripple rejection (Low pass) −3dB
BW1 = 1 / 2π (R4||R3) C3 = 1 / 2π (10k||10k) 100u = 0.318Hz

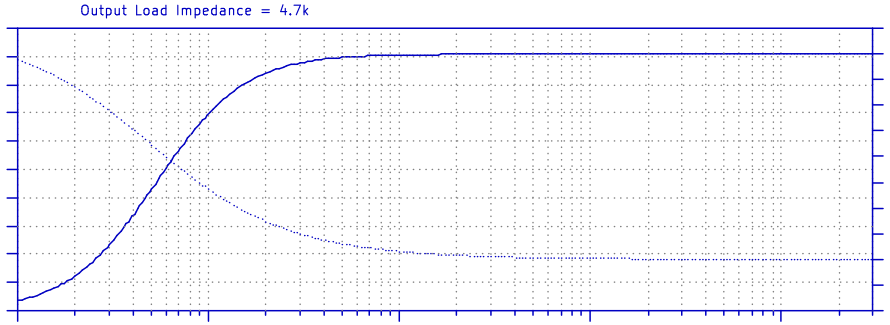
Input cut off (High pass) −3dB
BW2 = 1 / 2π R1 C1 = 1 / 2π 2.2k 10u = 7.95Hz

Output cut off (High pass) −3dB
BW3 = 1 / 2π R_{LOAD} C4 = 1 / 2π 4.7k 10u = 3.38Hz

GAIN = −R5/R1 = −20k/2.2k = −9.09

Resistor	Gain
None	9.09 (9.6dB)
150k	8.02 (9dB)
100k	7.58 (8.8dB)
68k	7.02 (8.5dB)
51k	6.53 (8.1dB)
36k	5.84 (7.7dB)
30k	5.45 (7.4dB)
24k	4.96 (7dB)
22k	4.76 (6.8dB)
20k	4.55 (6.6dB)
18k	4.31 (6.3dB)
16k	4.04 (6.1dB)
15k	3.90 (5.9dB)
13k	3.58 (5.5dB)
12k	3.41 (5.3dB)
11k	3.23 (5.1dB)
10k	3.03 (4.8dB)
9.1k	2.84 (4.5dB)
8.2k	2.64 (4.2dB)
7.5k	2.48 (3.9dB)
6.8k	2.31 (3.6dB)
6.2k	2.15 (3.3dB)
5.6k	1.99 (3dB)





https://www.analog.com/en/resources/app-notes/an-937.html

Sheet: /
File: ADA828_Amplifier.kicad_sch

Title: Generic AD828 Audio Amplifier Board

Size: A4	Date: Nov 2026	**Rev:**
KiCad E.D.A. 9.0.2		Id: 1/1